Regime-Conditional Tail Risk Modeling for Portfolio Risk Management

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Code: https://github.com/minh-stakc/algogator-tail-risk

1 Abstract

This paper investigates whether regime-aware risk models can produce better Conditional Value-at-Risk (CVaR) forecasts than the standard approaches portfolio managers typically use. Using daily returns from the S&P 500 and nine sector ETFs over 2005-2024, I detect market regimes with a Hidden Markov Model (HMM) and estimate tail risk within each regime using Peaks-over-Threshold extreme value theory. The results are mixed. Regime conditioning reveals something economically real - the high-volatility regime has a CVaR nearly three times larger than the calm regime, and cross-asset correlations jump sharply during stress. On standard out-of-sample backtesting, however, GARCH(1,1) still outperforms the regime-conditional model because it reacts to volatility changes one day at a time, while HMM regime detection lags sudden shocks by days or weeks. Regime models do excel during prolonged stress: in the 2022 rate-hike cycle, regime-conditional CVaR had only 0.4% violations versus 3-4% for all baselines. The practical takeaway is a two-layer framework - GARCH for daily risk limits and HMM regime probability as a strategic overlay for slower-moving macro risks.

2 Introduction

Most portfolio risk models assume return distributions stay roughly stable over time. That assumption works fine during normal markets, but it falls apart during crises - correlations spike, losses cluster, and distribution tails get much fatter. A model calibrated on mixed calm-and-crisis data systematically underestimates risk during the exact periods when accurate estimates matter most.

The metric this paper focuses on is Conditional Value-at-Risk (CVaR), also called Expected Shortfall - the expected loss given that losses exceed the 95th percentile. CVaR is preferred over VaR because it captures how bad losses actually get in the tail, not just where the tail starts.

Research Question

Do regime-aware risk models that combine Bayesian changepoint detection with dynamic dependence and extreme-value methods produce more reliable CVaR forecasts than traditional static or rolling-window models?

Hypothesis

Return distributions and cross-asset dependence structures behave differently across market regimes. Pooling observations from calm and stress periods leads to systematic underestimation of joint tail risk during crises. Separating estimation by regime should improve CVaR forecast accuracy and stability.

Approach

I detect regimes using a 2-state Gaussian HMM and a Bayesian changepoint algorithm (PELT), fit Generalized Pareto Distributions to the loss tails within each regime, and compare out-of-sample CVaR accuracy against three baselines: static historical simulation, rolling-window simulation, and GARCH(1,1).

3 Methodology

2.1 Data

I downloaded daily adjusted closing prices from Yahoo Finance for ten equity instruments: SPY (S&P 500) and nine sector SPDR ETFs (XLF, XLE, XLV, XLK, XLI, XLP, XLU, XLY, XLB), covering January 2005 through December 2024 - 5,031 trading days total. The sample intentionally spans the 2008 Global Financial Crisis, the 2020 COVID crash, and the 2022 Federal Reserve tightening cycle. VIX was also included as a supplementary stress indicator. Daily log returns were computed and an equal-weight portfolio constructed. Data were split into a training set (2005-2017, 3,271 days) and a test set (2018-2024, 1,760 days). Table 1 and Figure 1 summarize the portfolio's return properties.

Table 1: Portfolio return summary statistics (2005-2024)

Statistic	Value
Mean daily return	0.0356%
Std deviation	1.169%
Minimum (worst day)	-12.25%
Maximum (best day)	+10.62%
Excess kurtosis	14.14 (Normal = 0)
Skewness	-0.59

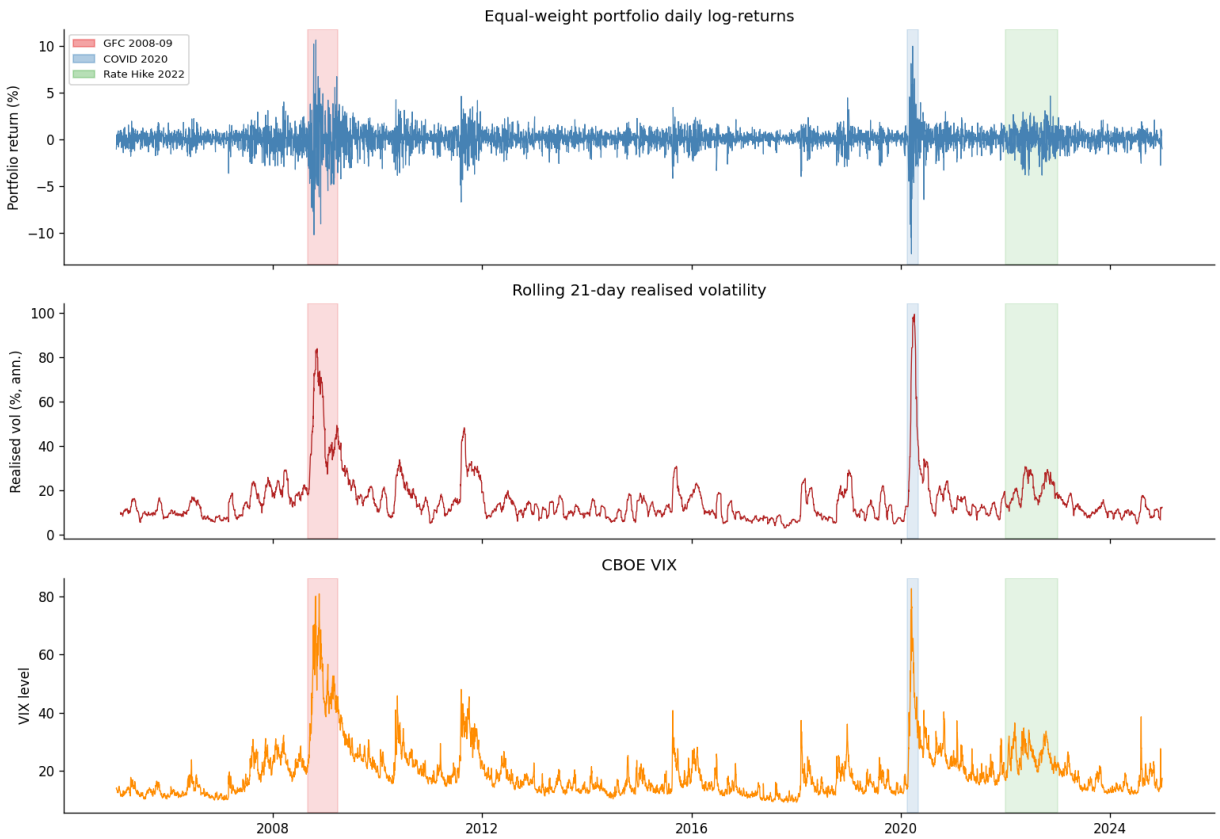

Figure 1: Portfolio returns, rolling volatility, and VIX (2005-2024). Shaded bands mark GFC 2008-09, COVID 2020, and Rate Hike 2022.

2.2 Regime Detection

Two complementary methods were used. The PELT algorithm with a BIC penalty was applied to rolling 21-day realized volatility, detecting 36 structural breakpoints - with key dates aligning to September 2008, February 2020, and May 2022 (Figure 3). A 2-state Gaussian HMM was then fitted to a feature matrix of rolling volatility, rolling mean return, and VIX level. The HMM produces both a soft posterior probability over states and a hard Viterbi assignment. Both states are highly persistent: P(stay low-vol) = 99.1%, P(stay high-vol) = 98.1% (Table 2, Figure 2).

Table 2: HMM estimated transition matrix

	To: Low-vol	To: High-vol
From: Low-vol	0.9911	0.0089
From: High-vol	0.0187	0.9813

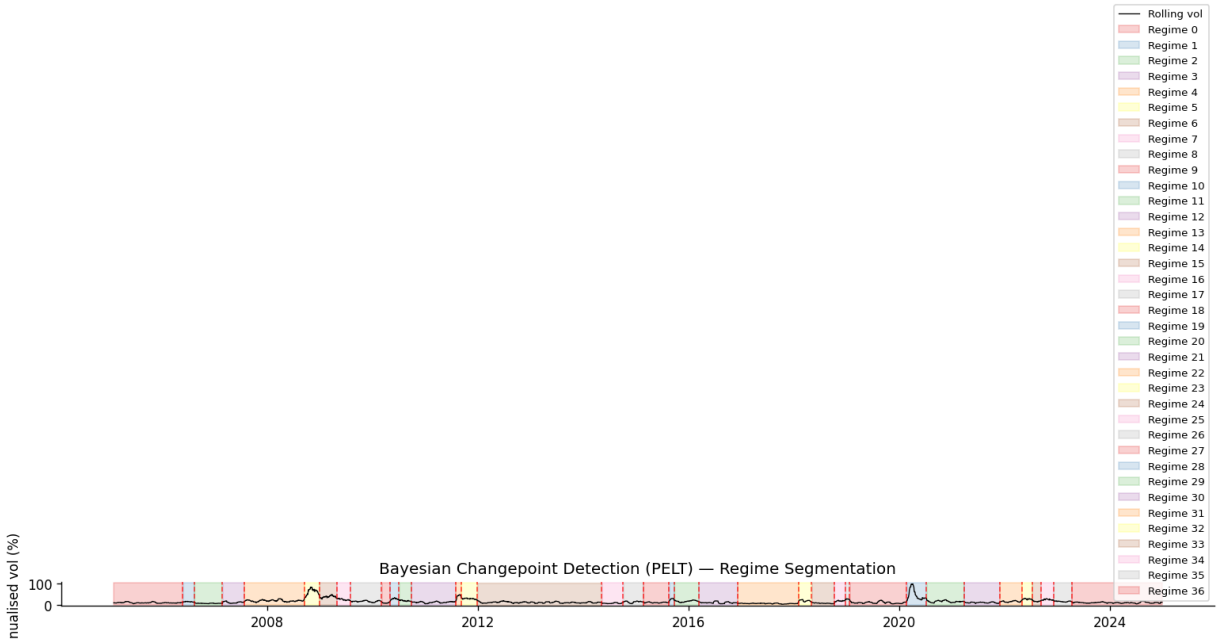

Figure 3: PELT changepoint segmentation with 36 detected structural breaks (red dashed lines).

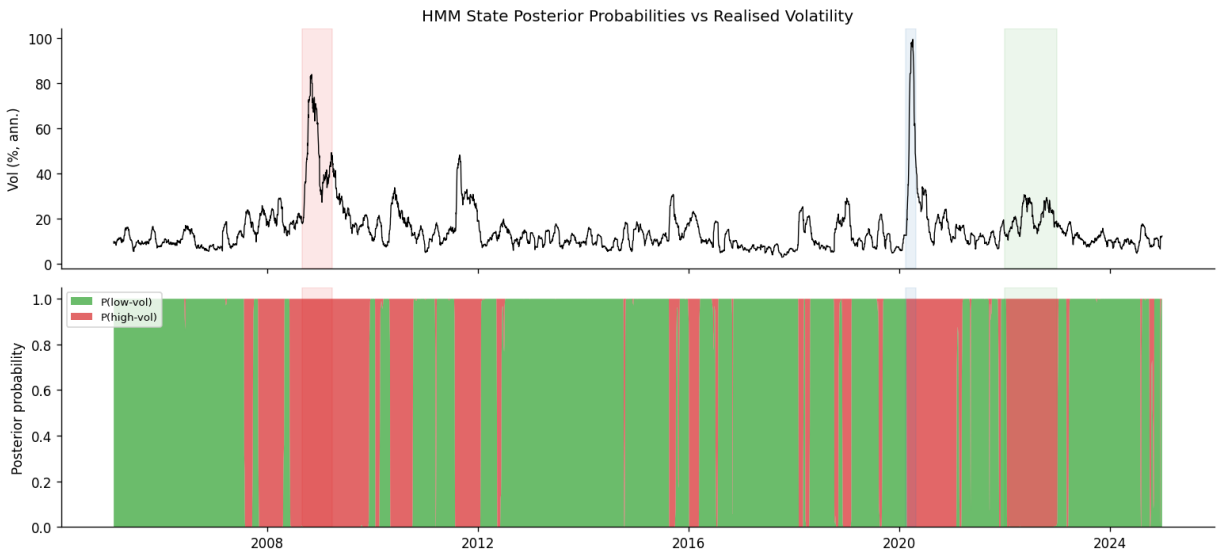

Figure 2: HMM posterior state probabilities. Green = P(low-vol), red = P(high-vol).

2.3 Tail Risk Modeling

For each detected regime, a Generalized Pareto Distribution (GPD) was fitted to loss exceedances above the 85th-percentile threshold using maximum likelihood. CVaR is computed analytically from the GPD shape parameter xi and scale parameter beta using the McNeil and Frey (2000) formula. Three baseline models were also estimated: Static Historical CVaR (full training sample, constant forecast), Rolling-Window CVaR (252-day sliding window), and GARCH(1,1) CVaR with normal innovations.

2.4 Copula Analysis

To model joint tail behavior across assets, I fitted a Gaussian copula (zero tail dependence

baseline), a Student-t copula with estimated degrees of freedom, and a bivariate Clayton copula capturing asymmetric lower-tail dependence between SPY and XLF.

2.5 Backtesting

Regime-conditional models were evaluated via an expanding-window walk-forward backtest with quarterly re-estimation and a 3-year minimum burn-in. Accuracy was assessed using the Kupiec (1995) Proportion-of-Failures test, the Christoffersen (1998) Conditional Coverage test, and pairwise Diebold-Mariano forecast comparison.

4 Results

4.1 Regime Characteristics

The HMM assigns 67.8% of days to the low-vol regime and 32.2% to the high-vol regime. Table 3 and Figure 4 show the key differences.

Table 3: Regime characteristics - HMM 2-state model (2005-2024)

Statistic	Low-vol Regime	High-vol Regime
Days (% of sample)	3,396 (67.8%)	1,615 (32.2%)
Ann. mean return	+16.54%	-6.81%
Ann. volatility	10.76%	28.77%
Excess kurtosis	1.39	6.05
Skewness	-0.345	-0.369
CVaR (95%)	1.56%	4.42%

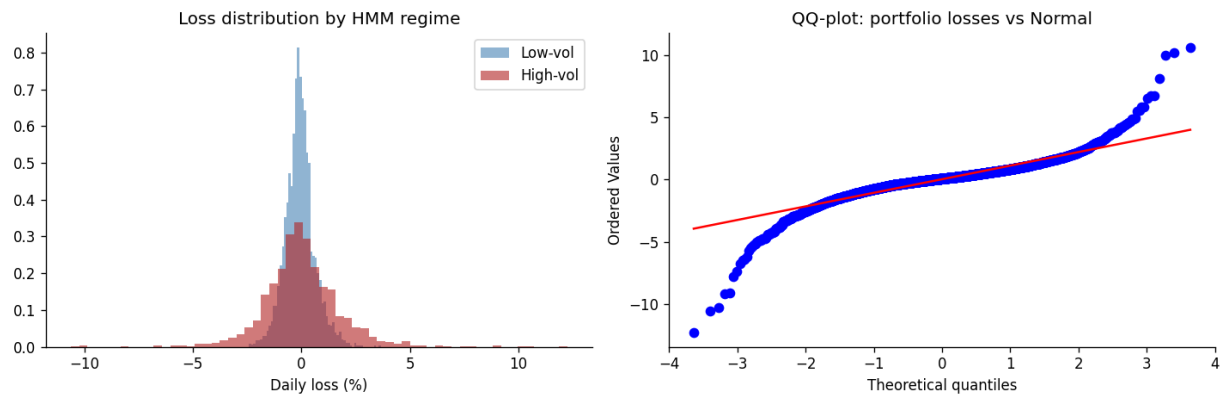

Figure 4: Loss distributions by HMM regime (left) and QQ-plot vs. Normal (right).

The high-vol regime produces CVaR roughly 2.8x larger (4.42% vs. 1.56%) and excess kurtosis over four times higher (6.05 vs. 1.39). Annualized returns are sharply negative (-6.8% vs. +16.5%). Figure 2 shows the HMM high-vol posterior probability spiking cleanly during GFC 2008, COVID 2020, and the 2022 rate-hike cycle.

4.2 Correlation Breakdown in Stress

Cross-asset correlations rise sharply in the high-vol regime. XLU (Utilities) is the most extreme: its correlation with SPY nearly doubles from 0.43 to 0.76. A static model calibrated on pooled data misses this entirely, understating joint tail risk precisely when it is highest.

Table 4: Cross-asset return correlation by regime (selected pairs)

Asset Pair	Low-vol	High-vol	Change
SPY - XLF	0.815	0.857	+0.042
SPY - XLE	0.580	0.790	+0.210
SPY - XLU	0.428	0.756	+0.328

SPY - XLV	0.741	0.859	+0.118
SPY - XLK	0.869	0.934	+0.065

4.3 EVT Results

Table 5 shows GPD parameters by regime. In the high-vol regime, xi = +0.116 confirms heavy Pareto-style tails. In the low-vol regime, xi = -0.073 indicates a bounded thin-tailed distribution. The pooled full-sample estimate (xi = +0.181) overstates tail heaviness in calm periods while understating it in stress. Figure 7 validates the GPD fit via a mean excess plot.

Table 5: GPD parameters by regime (training set 2005-2017)

Regime	n	xi (shape)	beta (scale)	CVaR 95%
Full sample	3,271	0.181	0.00847	2.92%
Low-vol	2,219	-0.073	0.00537	1.56%
High-vol	1,052	+0.116	0.01237	4.47%

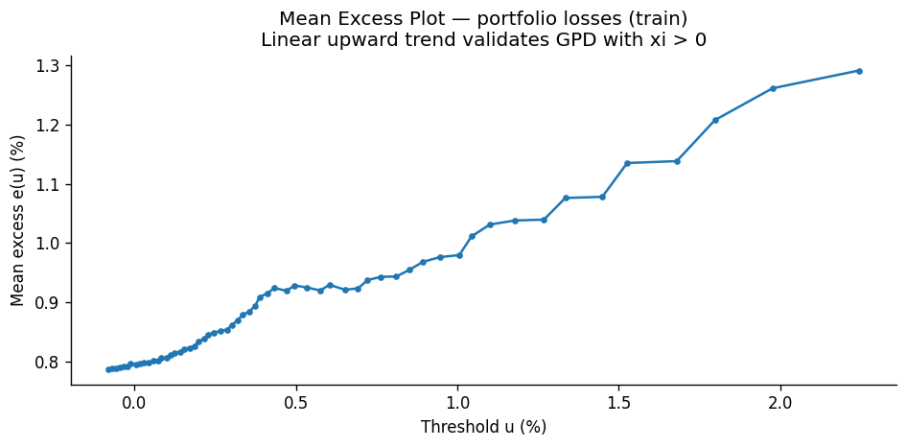

Figure 7: Mean excess plot confirming GPD fit validity (upward linear trend implies xi > 0).

4.4 Copula Analysis

The Student-t copula on all ten assets gives nu = 4.30, indicating substantial joint tail dependence. The Clayton copula on SPY-XLF gives lower tail dependence lambda = 0.770 - nearly 3 in 4 extreme SPY losses coincide with an extreme XLF loss. Tail dependence peaks during COVID 2020 (lambda = 0.694), confirming simultaneous crashes were especially likely during that episode (Table 6, Figure 5).

Table 6: Student-t copula tail dependence by stress period

Period	Est. nu	Bivariate lambda
Full sample (2005-2024)	4.30	0.543
GFC 2008-09	7.98	0.458
COVID 2020	6.69	0.694
Rate Hike 2022	10.85	0.429

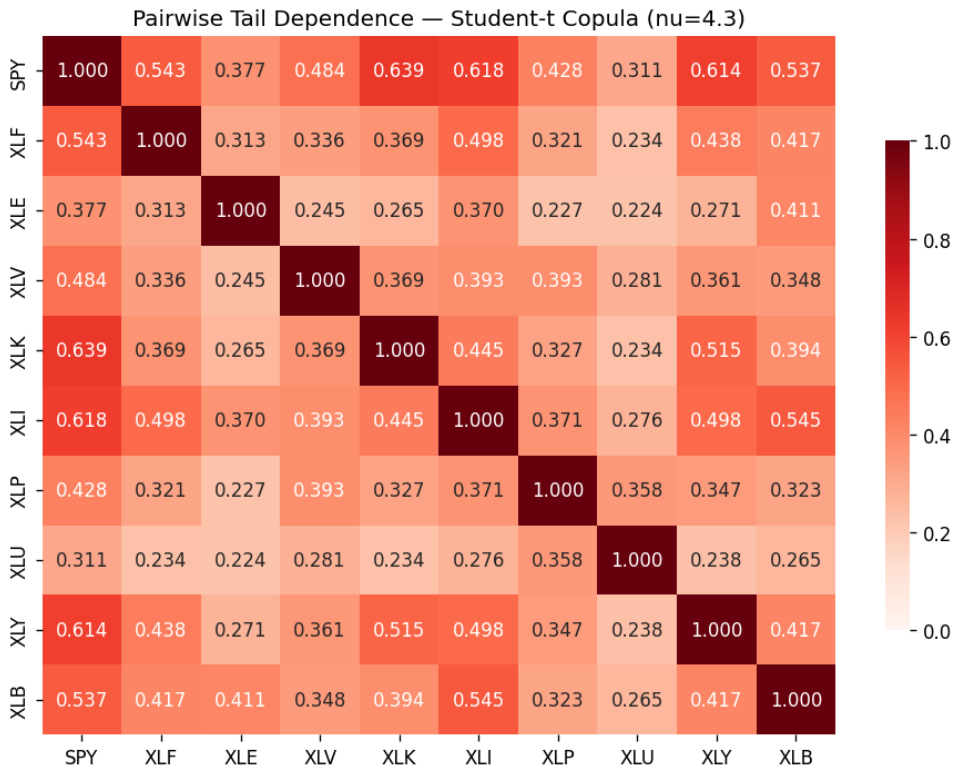

Figure 5: Pairwise tail dependence heatmap - Student-t copula (nu = 4.30).

4.5 Out-of-Sample Backtesting

Table 7 shows the full backtest over the 2018-2024 test period. A properly calibrated 95% CVaR model should produce a 5% violation rate.

Table 7: CVaR model comparison - out-of-sample test period 2018-2024

Model	Viol Rate	Expected	Kupiec p	CC p	MAE
Static-Hist	1.48%	5.0%	<0.001*	<0.001*	0.0300
Rolling-252	2.67%	5.0%	<0.001*	<0.001*	0.0274
GARCH	3.58%	5.0%	0.004*	0.009*	0.0205
Regime-Hist	3.30%	5.0%	<0.001*	<0.001*	0.0280
Regime-EVT	3.24%	5.0%	<0.001*	<0.001*	0.0282

** Rejects H0 (violation rate = 5%) at 5% significance.*

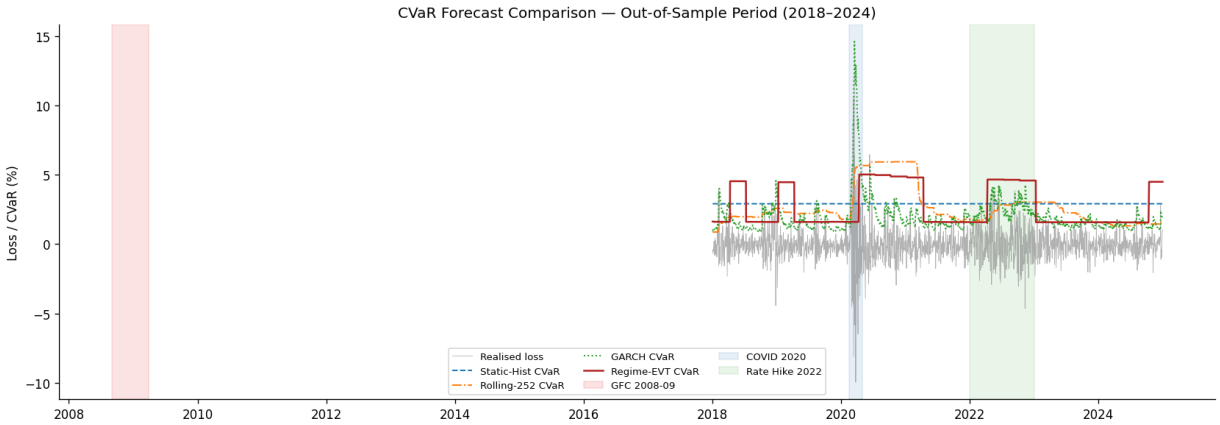

Figure 6: CVaR forecast time series for all five models, 2018-2024 test period.

No model hits 5%. Static-Hist is the worst at 1.48% - overstating risk and tying up unnecessary capital. GARCH comes closest (3.58%) with the lowest MAE (0.021). Diebold-Mariano tests confirm GARCH beats Regime-EVT on squared forecast error (DM = 5.71, $p < 0.001$).

4.6 Stress-Period Analysis

Table 8 and Figure 8 break down violation rates during the named stress events.

Table 8: CVaR violation rates during named stress periods

Model	Full Sample	COVID 2020	Rate Hike 2022
Static-Hist	1.48%	25.00%	3.19%
Rolling-252	2.67%	17.31%	4.38%
GARCH	3.58%	11.54%	3.59%
Regime-Hist	3.30%	28.85%	0.40%
Regime-EVT	3.24%	28.85%	0.40%

Figure 8: CVaR violation days (red dots) per model, 2018-2024 test period.

The split is stark. During COVID 2020 regime models perform worst (28.9% violations) - the HMM was still in low-vol mode when the crash started. During the 2022 rate-hike cycle, regime models are best by far (0.4% violations vs. 3-4% for baselines). Once the HMM locked onto the high-vol state, it held it through the entire cycle.

5 Discussion

5.1 What the Results Mean

The hypothesis gets partial support. Regime conditioning clearly captures something real - tail distributions, correlations, and joint tail dependence all differ significantly across regimes. Any risk framework that ignores this structure is missing important information. That said, the regime detection lag is the core problem: by the time the HMM is confident the regime has shifted, the worst losses may already be behind you. GARCH avoids this because it responds to yesterday's realized return directly, adapting at a one-day lag instead of a multi-week lag.

5.2 Implications for the Fund

The most practical takeaway is a two-layer framework. Use GARCH for daily CVaR limits and position sizing - it is the best single tool for day-to-day risk management. Layer HMM regime probability on top as a strategic overlay: when P(high-vol) crosses ~40%, scale back exposure pre-emptively before volatility fully materializes. The 2022 result shows regime models are very effective once a sustained stress environment is established. The copula analysis also informs hedging - when tail dependence spikes, apparent diversifiers stop providing protection.

5.3 Limitations

The regime detection lag is the main limitation. A forward-looking indicator - credit spreads or VIX term structure slope - could help reduce it. The equal-weight portfolio simplifies the analysis; a real fund portfolio with time-varying weights would require CVaR computed on the actual weight vector. Copula-based CVaR was estimated in-sample only, and PELT was applied to the full sample for visualization - a live implementation would need BOCPD.

6 Conclusion

Market regimes genuinely matter for tail risk. The high-vol regime has fundamentally different tail properties, higher correlations, and greater joint crash risk - mixing the two in one model distorts estimates in both directions. But the regime detection lag prevents this from translating into better out-of-sample CVaR numbers during sudden shocks like COVID 2020. GARCH handles those better.

The regime approach is not a replacement for GARCH - it is a complement. As a strategic overlay, it flags elevated tail dependence before a sustained stress cycle fully materializes and delivers significantly better CVaR estimates during slow-moving macro stress like 2022. The recommended approach for the fund is to run both in parallel - GARCH for daily limits, HMM regime probability for strategic capital allocation decisions.

Future Work

The most important extension is incorporating forward-looking signals (credit spreads, VIX term structure) to reduce the regime detection lag. Fitting regime-specific copulas and running a proper walk-forward copula backtest would also strengthen the multivariate tail analysis.

7 References

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8 Appendices

All figures are embedded inline within their respective sections. Source code, data pipeline, and model implementations are available at: https://github.com/minh-stakc/algogator-tail-risk

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